

cloning or imaging).

- Copying the files manually and adjusting the configuration.
- Making use of some automated tools.

Each method has its own pros and cons. I prefer the second method. However, we'll have a quick review of each one and many of the things explained here will be useful, regardless of which one you choose.

Which one to choose?

As always, it depends on your needs. When it comes to cloning, nothing is better than the imaging method, as the disk is copied bit-by-bit, preserving the low-level details like file system structures. Disk images have other wonderful applications also. One is software-based data recovery, where a tool gets access to each bit of the disk without actually hurting the dying disk. Disk cloning can be done using the standard GNU/Linux utilities like *cp*

and *dd*, and it takes just a single line of command. However, this kind of verbatim copying is an overhead for normal purposes, because it'll copy the junk data in the blank areas also. Besides, it may not work if the source and destination are of different sizes. Another issue is that applications will get confused if you keep both the original and the cloned disks connected or mounted at the same time, because some parts of disks like UUIDs are expected to be unique.

Copying files and reconfiguring them is the most flexible method. It allows you to choose a target partition of a different size and file system type. You can omit unwanted files. The only overhead is to adjust the configurations, which isn't that hard.

Automated tools include *Partclone* and *Clonezilla live*, a bootable GNU/Linux distribution with *Partclone* pre-installed. I haven't tested any of these automatic tools, since the manual methods are already simple; and even if

you use automatic cloning tools, you'll have something to fix manually.

Now you know why the second method is preferred if you just want to get your existing OS installation copied on a new disk or machine. To make things clear, a side-by-side comparison is given in Table 1.

Overview of manual copying and tweaking

Now that we've decided to perform manual copying and reconfiguration, this is the algorithm that should be followed.

1. Boot into a live GNU/Linux CD or USB stick, or a secondary installation that you already have.
2. Know your machine – find the device names, right partitions, etc.
3. Remove junk files and applications.
4. Partition and format, if necessary.
5. Copy the files.
6. Edit the configuration files.
7. Fix the bootloader.

Table 1: Comparing various OS cloning methods

Disk cloning	Copying files and reconfiguring	Automatic tools
Can be used to copy an entire disk or a partition.	Can be used to copy the selected contents of a partition.	Scope depends on the tool.
Pros: • Takes just a single line of command. • You get an exact copy, including the partitions and the unlinked (removed) data. • Best choice for software-based data recovery. • Fixing the bootloader is necessary only if you are cloning a partition, rather than the whole disk.	Pros: • Free space is not copied and this saves time. • Source and destination disks can be of different sizes, layouts and/or file system types. • You can omit large portions (e.g.: <i>/home</i>), if you want. • You can use your favourite file browser to copy the files, and there's less chance that you'll pick the wrong target.	Pros: • Easy to use. • Some packages come as live (bootable) systems with pre-installed utilities. • May support features like free-space skipping.
Cons: • Cloning the junk data in the free space is an overhead and takes time. • UUIDs get cloned too, causing confusion or even crashes. • No opportunity to change the file system type. • Can be dangerous if you chose an incorrect target.	Cons: • Tables like the <i>fstab</i> need changes. • The bootloader has to be installed or updated. • File metadata can get corrupted if the wrong command is used for copying, or the target partition is of a different (limited) file system type.	Cons: • Might perform tasks automatically that you don't want. • Sometimes you'll still have to do something manually (like fixing the bootloader).